

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, data pipelines, and fintech developer tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
- Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails, Semantic Retrieval

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, Amazon Kinesis, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, SQLite, DuckDB, Polars, Parquet

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, Serverless Architecture, AWS Lambda, ECS, EKS, Vercel

PROJECTS

- AI Agent Harness Configs** - Python/TypeScript MCP workflow catalog with 40+ reusable skills, FastMCP configs, and cross-runtime agent automation.
- Built 40+ instruction and skill packs plus harness configs across Claude Code, Codex, GitHub Copilot, Cursor, and related IDE agents.
 - Shipped generated docs, install flows, and portable tool and MCP presets for code review, security, infrastructure, research, and workflow automation.

- NBA Basketball Database** - Python warehouse using Pandas, Polars, SQLAlchemy, DuckDB, and Pandera across 65K+ games, 13M+ events, and Parquet/SQLite exports.
- Implemented Pandera-validated NBA API ETL with daily warehouse updates plus DuckDB, SQLite, Parquet, and CSV exports.
 - Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

- ProxyWhirl** - Python/SQLite proxy toolkit with nine rotation strategies, 100+ sources, PyPI distribution, and self-healing health checks.
- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
 - Built self-healing validation for 100+ proxy sources with circuit breakers, auto-ejection, and recovery promotion.

- Personal Website: w4w.dev** - Next.js/React/TypeScript publishing system with Tailwind CSS, MDX, durable content, SEO, and polished public UI.
- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
 - Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)
Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate

IBM | Feb 2021